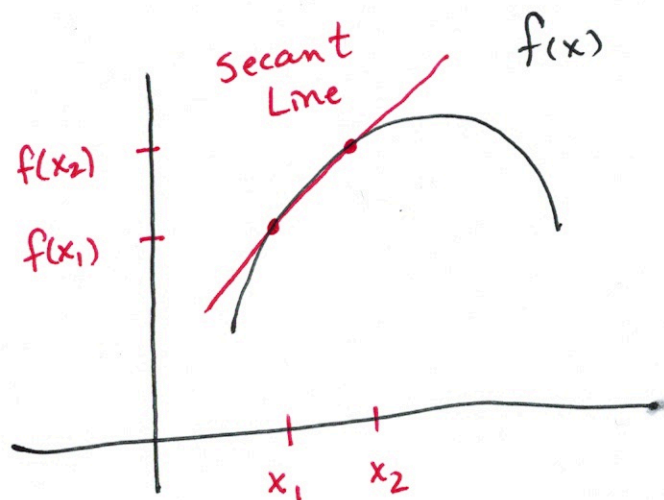
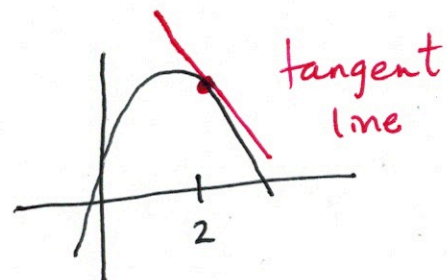


## Section 2.1 Secant Lines and Velocity



$$m = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



Ex ①  $f(x) = -x^2 + 2x + 3$

Want slope at  $(2, 3)$

use  $x = 1.9, 2.1, 1.99, 2.01$

$x = 1.9$   $\rightarrow y = 3.19$

$$m = \frac{3.19 - 3}{1.9 - 2} = \frac{0.19}{-0.1} = -1.9$$

$x = 2.1$   $\rightarrow y = 2.79$

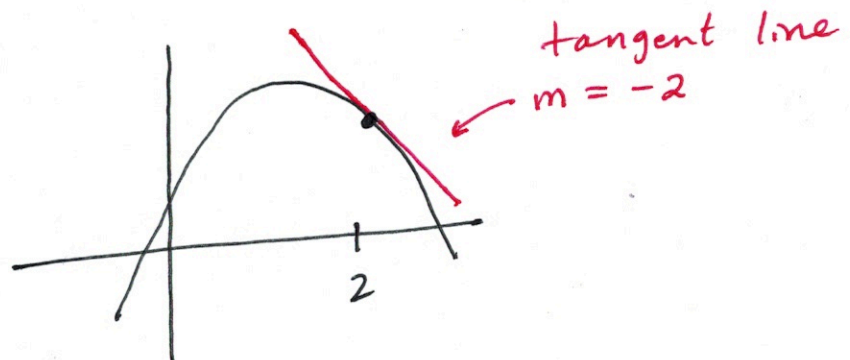
$$m = \frac{2.79 - 3}{2.1 - 2} = \frac{-0.21}{0.1} = -2.1$$

Ex ①  $x = 1.99 \rightarrow y = 3.02$   $(2, 3)$

$$m = \frac{3.02 - 3}{1.99 - 2} = \frac{0.02}{-.01} = -2 \checkmark$$

$x = 2.01$   $\rightarrow y = 2.98$

$$m = \frac{2.98 - 3}{2.01 - 2} = \frac{-0.02}{.01} = -2 \checkmark$$



Ex (2)  $h(t) = -16t^2 + 80t + 6$

is height in feet after  $t$  seconds.

Find average velocity on the  
interval  $[t_1, t_2]$   $[1.9, 2.1]$

find slope of  
the secant line

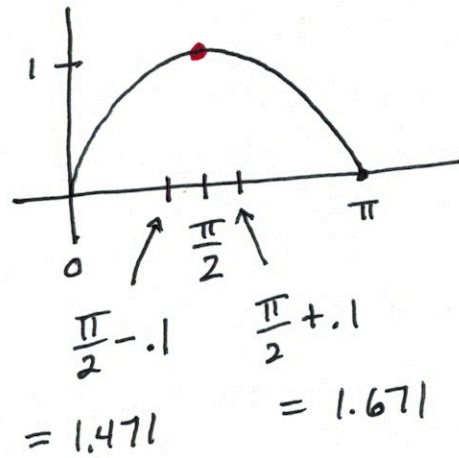
$$m = \frac{h(2.1) - h(1.9)}{2.1 - 1.9}$$

$$m = \frac{(103.44 - 100.24)}{0.2}$$

$$m = 16 \frac{\text{ft}}{\text{sec}}$$

Ex (3)  $f(x) = \sin x$

Estimate slope of the tangent  
line when  $x = \frac{\pi}{2}$



$$m = \frac{\sin(1.671) - \sin(1.471)}{1.671 - 1.471}$$

$$m = -0.2$$

Ex (4)

| hour<br>$t$ | miles |
|-------------|-------|
| 1           | 48    |
| 2           | 105   |
| 3           | 165   |
| 4           | 212   |

Estimate velocity  
over each 1 hour  
interval.

$$\underline{1^{\text{st}} \text{ hour}} \quad \frac{48 \text{ miles}}{1 \text{ hour}} = 48 \text{ mph}$$

$$\underline{2^{\text{nd}} \text{ hour}} \quad \frac{105 - 48}{2 - 1} = 57 \text{ mph}$$

$$\underline{3^{\text{rd}} \text{ hour}} \quad \frac{165 - 105}{3 - 2} = 60 \text{ mph}$$

$$\underline{4^{\text{th}} \text{ hour}} \quad \frac{212 - 165}{4 - 3} = 47 \text{ mph}$$